

Study-Material

Operating Systems Study Guide

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Chapter 1: Introduction to Operating Systems

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Key Concepts

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- **Definition**: An operating system is a program that controls execution of all other programs

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- **Role**: Acts as intermediary between users and computer hardware

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- **Objectives**: Convenience, efficiency, extensibility

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- **Evolution**: From open shop → batch systems → multiprogramming → time-sharing → modern systems

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Important Definitions

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- **Operating System (OS)**: Controls all execution, multiplexes resources, abstracts complexity

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- **Kernel**: Core part of OS (approximately)

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- **Process**: A program in execution (dynamic vs static program)

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Examples

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- ****Historical Evolution****:

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- 1949: EDSAC first stored-program machine

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- 1950s: "Open Shop" with operator-mediated job submission

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- Batch systems with tape drives and resident monitors

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- Multiprogramming with memory caching

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- Time-sharing systems (CTSS, Unix, VMS)

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Exam Tips

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- Understand the fundamental role of an OS beyond just running programs

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- Know the progression of OS evolution and why each step was necessary

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- Remember that OS $\approx$ kernel for exam purposes

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Practice Questions

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1. Explain the objectives of an operating system.

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2. Describe the evolution from open shop systems to time-sharing systems.

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3. What is the difference between a program and a process?

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Chapter 2: Processes & Scheduling

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Key Concepts

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- **Process States**: New, Ready, Running, Blocked, Exit

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- **Context Switching**: Saving and restoring process state

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- **CPU Scheduling Algorithms**: FCFS, SJF, SRTF, Round Robin

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- **Scheduling Criteria**: CPU utilization, throughput, turnaround time, waiting time, response time

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Important Definitions

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- **Process Control Block (PCB)**: Data structure maintained by OS with process information

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- **Context**: Machine environment including program counter, registers, status flags

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- **Convoi Effect**: Poor performance in FCFS when long process arrives first

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- **Time-Slice/Quantum**: Fixed unit of time for Round Robin scheduling

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Examples

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FCFS Scheduling:

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Process | Burst Time | Arrival Order 1,2,3

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P1 | 24 | Waiting: 0, 24, 28

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P2 | 3 | Average: 17.33

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P3 | 3 |

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****SJF Optimal Example**:**

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Process | Arrival | Burst | Completion | Waiting

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P1 | 0 | 7 | 7 | 0

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P2 | 2 | 4 | 11 | 5

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P3 | 4 | 1 | 12 | 8

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P4 | 5 | 4 | 16 | 12

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Average Waiting Time: $(0+5+8+12)/4 = 6.25$

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...

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****SRTF Preemptive Example**:**

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Process | Arrival | Burst

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P1 | 0 | 7

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P2 | 2 | 4

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P3 | 4 | 1

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P4 | 5 | 4

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Gantt Chart: P1(0-4) → P3(4-5) → P2(5-9) → P4(9-13) → P1(13-16)

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Waiting Times: P1=9, P2=1, P3=0, P4=2 → Avg = 3

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Exam Tips

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- **Memory**: Know Gantt chart construction for each algorithm

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- **Comparison**: SJF minimizes average waiting time but requires burst time prediction

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- **Trade-offs**: FCFS simple but suffers from convoy effect; SRTF optimal but complex

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- **Prediction Formula**: $\tau_{n+1} = \alpha t_n + (1-\alpha)\tau_n$

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Practice Questions

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1. Compare FCFS and SJF scheduling. When does each perform better?

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2. Calculate waiting times for the SRTF example above.

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3. Explain how exponential averaging can predict burst times.

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4. What are the advantages and disadvantages of Round Robin scheduling?

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Chapter 3: Memory Management

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Key Concepts

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- **Virtual Memory**: Provides each process with own address space

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- **Protection**: Base and limit registers prevent unauthorized access

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- **Memory Management Schemes**: Segmentation vs Paging

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Important Definitions

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- **Address Space**: Virtual memory space allocated to a process

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- **Base Register**: Starting address of process in physical memory

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- **Limit Register**: Size of process's memory allocation

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- **Segmentation**: Memory management using variable-sized segments

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- **Paging**: Memory management using fixed-size pages

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Examples

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****Base/Limit Protection**:**

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Physical Memory: 0x0000 to 0xFFFF

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Job 1: Base=0x0000, Limit=0x1000 (accesses 0x0000-0x0FFF)

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Job 2: Base=0x1000, Limit=0x2000 (accesses 0x1000-0x2FFF)

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Exam Tips

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- Understand memory protection hardware mechanism

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- Know difference between segmentation and paging

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- Remember that only kernel mode can modify base/limit registers

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Practice Questions

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1. Explain how base and limit registers provide memory protection.

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2. Compare segmentation and paging memory management schemes.

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3. What happens when a process tries to access memory outside its limits?

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Chapter 4: I/O & Device Management

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Key Concepts

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- **Device Management**: Controlling and coordinating I/O devices

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- **Interrupt-Driven I/O**: Efficient handling of device operations

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- **Spooling**: Using disk as cache for I/O operations

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Important Definitions

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- **Spooling**: Simultaneous Peripheral Operations Online - buffering I/O operations

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- **Interrupt**: Hardware signal requesting immediate attention from CPU

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Examples

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- **Spooling System**: Tape input cached on disk, allowing overlap of I/O operations

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- **Interrupt Handling**: Device signals completion, CPU handles other tasks meanwhile

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Exam Tips

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- Understand the role of interrupts in efficient I/O processing

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- Know how spooling improves system performance

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Practice Questions

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1. Explain the benefits of spooling systems over direct I/O.

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2. How do interrupts improve I/O efficiency?

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Chapter 5: Protection

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Key Concepts

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- **Dual-Mode Operation**: User mode vs Kernel mode

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- **Privileged Instructions**: Operations only allowed in kernel mode

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- **System Calls**: Secure way for applications to invoke OS services

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Important Definitions

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- **User Mode**: Execution mode for application programs

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- **Kernel Mode**: Execution mode for operating system

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- **Mode Bit**: Hardware flag distinguishing user/kernel mode

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- **System Call**: Interface between applications and operating system

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Examples

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****Mode Switching****:

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...

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Application → System Call → Kernel Mode → OS Service → Return to Application

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...

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Exam Tips

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- Remember that certain instructions are privileged

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- Understand the role of system calls in providing OS services

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Practice Questions

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1. Explain dual-mode operation and its importance.

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2. What are system calls and how do they enable application-OS interaction?

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Chapter 6: Filing Systems

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Key Concepts

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- **File Organization**: Hierarchical directory structures

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- **Storage Allocation**: Methods for organizing disk space

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- **Security and Protection**: Access control mechanisms

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Important Definitions

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- **File**: Sequence of bytes with attributes and protection information

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- **Directory**: Collection of files and subdirectories

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Examples

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- **Hierarchical File System**: Files organized in directories and subdirectories

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Exam Tips

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- Understand basic file system concepts and organization

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Practice Questions

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1. Describe the structure of a hierarchical file system.

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2. What are the key components of file system protection?

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Chapter 7: Unix Case Study

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Key Concepts

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- **Unix Philosophy**: Simple, modular design

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- **Process Management**: fork() and execve() system calls

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- **File System**: Everything is a file

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Important Definitions

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- **fork()**: Creates child process as clone of parent

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- **execve()**: Loads new program into process memory

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Examples

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Unix Process Creation:

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```
```c
```

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```
pid = fork(); // Create child process
```

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```
if (pid == 0) {
```

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```
execve("/bin/lis", argv, envp); // Child executes new program
```

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```
} else {
```

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```
wait(NULL); // Parent waits for child
```

*Generated by StudyMate*

```
}
```

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...

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### Exam Tips

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- Know key Unix system calls for process management

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- Understand Unix's approach to modularity

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### Practice Questions

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1. Explain the Unix fork-exec process creation model.

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2. What are the advantages of Unix's "everything is a file" philosophy?

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## Chapter 8: Windows NT Case Study

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### Key Concepts

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- **Hybrid Kernel**: Blends kernel and microkernel approaches

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- **Process Management**: CreateProcess() API

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- **Security Model**: Integrated security and protection

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### Important Definitions

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- **CreateProcess()**: Windows API for process creation

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- **Hybrid Kernel**: Microkernel with performance-critical services in kernel

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### Examples

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**Windows Process Creation**:

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```c

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CreateProcess("program.exe", command_line,

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NULL, NULL, FALSE, 0,

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NULL, NULL, &si, π);

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...

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Exam Tips

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- Compare Unix and Windows NT approaches

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- Understand Windows' hybrid kernel design

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Practice Questions

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1. Compare process creation in Unix and Windows NT.

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2. What are the advantages of Windows NT's hybrid kernel approach?

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Comprehensive Exam Tips

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High-Yield Topics

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1. **CPU Scheduling Algorithms**:

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- Know Gantt charts for FCFS, SJF, SRTF, Round Robin

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- Understand trade-offs between algorithms

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- Memorize prediction formula: $\tau_{n+1} = \alpha t_n + (1-\alpha)\tau_n$

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2. **Process Management**:

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- Process states and transitions

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- Context switching overhead

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- fork-exec model (Unix) vs CreateProcess (Windows)

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3. **Memory Protection**:

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- Base/limit register mechanism

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- Segmentation vs paging differences

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4. **Operating System Structures**:

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- Monolithic vs Microkernel vs Hybrid

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- System calls and protection mechanisms

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Formula Sheet

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- **Exponential Averaging**: $\tau_{n+1} = \alpha t_n + (1-\alpha)\tau_n$

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- **Average Waiting Time**: Sum of individual waiting times / number of processes

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- **CPU Utilization**: Time CPU is doing useful work / total time

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Comparison Points for Exams

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| Aspect | Unix | Windows NT |

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|-----|-----|-----|

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| Process Creation | fork() + execve() | CreateProcess() |

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| Kernel Type | Monolithic | Hybrid |

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| Philosophy | Everything is a file | Integrated security |

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Final Practice Questions

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1. **Short Answer**: Explain why time-sharing systems were developed.

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2. **Calculation**: Given processes with burst times 10, 1, 4, 2ms arriving at times 0, 2, 4, 6ms respectively, calculate average waiting time using SRTF.

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3. **Comparison**: Compare monolithic and microkernel architectures. Which performs better and why?

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4. **Application**: A system has 5 processes with varying CPU bursts. Which scheduling algorithm would minimize average waiting time?

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5. **Design**: Why is dual-mode operation necessary in modern operating systems?

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This study guide covers all major topics from the course material. Focus on understanding the concepts rather than memorizing details, and practice the calculation-based questions thoroughly.

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